

# PHD21 Computation methods: Replication report

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**Note.** I have picked two models which constitute the building blocks of my first-year project. In this context, I put more weight on the intensive rather than the extensive margin of the exercise. I use Julia and teach myself all the key “tricks” to speed up the computation process, prioritising precision and replicability. In the case of Buera et al. (2015), who provide no replication package and give scant details on the values of the model’s moments, I remain as faithful to the author’s described methods. In so doing, I like to think that I conduct a “stress test”, which I believe the paper does not fully pass. When it comes to Elsby and Michaels (2013), which is more widely used and replicated, I translate the model to Julia and use the repeated transition method of Lee (2025) to improve on the authors’ current methodology (Krusell and Smith, 1998).

## 1 Elsby and Michaels (2013): Marginal jobs, heterogenous firms

## 2 Buera et al. (2015): Anatomy of a credit shock

### 2.1 Overall impression

Having initially treated the paper as a potential “easy win,” I have been humbled by the model’s fragility. The authors’ results cannot be fully replicated, which I believe to be mostly attributable to the framework being extremely sensitive to minor changes to the productivity process. I also suspect that Buera et al. (2015) use more tricks to find the steady state solution reported in the paper, especially given that the described procedure is a baseline approach to solve the Aiyagari (1994) class of models.

### 2.2 Framework

The model features agents who are heterogeneous along two dimensions, their wealth ( $a$ ) and entrepreneurial productivity ( $z$ ). If an agent chooses to be an entrepreneur, she hires  $l$  workers. If she remains a worker, she earns wage  $w_t$ , irrespective of her employment status. Workers can be unemployed, with the job destruction endogenously determined by the downsizing entrepreneurs and the job finding rate, the latter of which is determined by the number of workers and job openings.

The value function for agent  $i \in \{e, w\}$  is:

$$\begin{aligned} V^i(a, z) = & \max_{a', c} \{ u(c) + \beta \mathbb{E}[V(a', z)] \} \\ \text{s.t } & c + a' = (1 + r)a + \Pi(a, z) \mathbb{1}\{i = e\} + w_t \mathbb{1}\{i = w\} - \tau, \end{aligned} \tag{1}$$

where  $u(c)$  is the utility of consumption  $c$ ,  $\beta$  is the discount rate,  $r$  is the interest rate,  $\Pi(a, z)$  is the profit level, and  $\tau$  is the tax financing the unemployment benefit (which is set to be the same as the wage).

Entrepreneur  $(a, z)$  earns the following profit:

$$\begin{aligned}\Pi(a, z) = \max_{k, l} \{ & Azk^\alpha l^\theta - wl - (r + \delta)k \} \\ \text{s.t } & k = \max\{k^*, \lambda a\},\end{aligned}\tag{2}$$

## 2.3 Steady state

Algorithm 1 fleshes out Buera et al. (2015)'s steady state solution approach. While pedagogically clean, the method is relatively inefficient – I keep using it to remain faithful to the paper. To fully test the model's performance, I allow for two methods of solving the value function iteration routine. First, I search for a solution on the grid, which is set to have 40 and 400 points for productivity and assets respectively.<sup>1</sup> This is fast, but potentially less accurate, with the VFI routine frequently stalling for lower asset grid sizes. Second, I search for a solution using the interpolated values (for both  $V^e$  and  $V^w$  to avoid smoothing over the kink).

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<sup>1</sup>Buera et al. (2015) only specify the size of the productivity grid, which I keep unchanged in all experiments.

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**Algorithm 1:** Stationary equilibrium (Buera et al., 2015)

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**Input:** Parameters  $(\sigma, \beta, \alpha, \theta, \delta, \psi, \eta, \lambda)$ , grids  $(\mathcal{A}, \mathcal{Z}, \mathcal{L})$ , tolerances  $(\delta_r, \delta_\tau, \delta_w, \delta_v, \delta_d)$

**Output:** Prices  $(r^*, w^*, \tau^*)$ , policies  $(c, a', k, l, o)$ , distribution  $G^*$ , unemployment  $U^*$

```
1 Initialize  $r^{(0)}$ ;
2 repeat
3   Initialize  $\tau^{(0)}$ ;
4   repeat
5     Initialize  $w^{(0)}$ ;
6     repeat
7       // Value function iteration
8       repeat
9         Compute  $v^W(a, z)$  from Equation (1);
10        Compute  $v^E(a, z)$  from Equation (1) using static FOCs for  $(k, l)$  s.t.
11           $k \leq \lambda a$  from Equation (2);
12          Set  $v(a, z) = \max\{v^W, v^E\}$ ; record  $o(a, z), a'(a, z)$ ;
13        until  $\|V^{n+1} - V^n\|_\infty < \varepsilon_v$ ;
14        // Forward simulation
15        Iterate  $G_{n+1} = \mathcal{T}(G_n; a', o, \psi, \mu)$  until  $\|G_{n+1} - G_n\| < \varepsilon_G$ ;
16        Compute  $L^d = \int l dG, L^s = 1 - \int \mathbb{1}\{o = E\} dG - U$ ;
17        Update  $w$  by bisection;
18        until  $|L^d - L^s| < \varepsilon_w$ ;
19        Compute  $U$ , set  $\tau^{\text{new}} = wU$ ;
20        Update  $\tau \leftarrow \eta_\tau \tau^{\text{new}} + (1 - \eta_\tau)\tau$ ;
21      until  $|\tau^{\text{new}} - \tau| < \varepsilon_\tau$ ;
22    Compute  $K^d = \int k dG, K^s = \int a dG$ ;
23    Update  $r$ , raising if excess demand and lowering if excess supply. ;
24  until  $|K^d - K^s| < \varepsilon_r$ ;
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## References

- Aiyagari, S. R. (Aug. 1994). “Uninsured Idiosyncratic Risk and Aggregate Saving”. en. In: *The Quarterly Journal of Economics* 109.3, pp. 659–684. ISSN: 0033-5533, 1531-4650. DOI: [10.2307/2118417](https://academic.oup.com/qje/article-lookup/doi/10.2307/2118417). URL: <https://academic.oup.com/qje/article-lookup/doi/10.2307/2118417> (visited on 10/17/2025).
- Buera, Francisco J., Roberto N. Fattal Jaef, and Yongseok Shin (Jan. 2015). “Anatomy of a credit crunch: From capital to labor markets”. en. In: *Review of Economic Dynamics* 18.1, pp. 101–117. ISSN: 10942025. DOI: [10.1016/j.red.2014.11.001](https://linkinghub.elsevier.com/retrieve/pii/S1094202514000714). URL: <https://linkinghub.elsevier.com/retrieve/pii/S1094202514000714> (visited on 04/17/2026).
- Elsby, Michael W. L and Ryan Michaels (Jan. 2013). “Marginal Jobs, Heterogeneous Firms, and Unemployment Flows”. en. In: *American Economic Journal: Macroeconomics* 5.1, pp. 1–48. ISSN: 1945-7707, 1945-7715. DOI: [10.1257/mac.5.1.1](https://pubs.aeaweb.org/doi/10.1257/mac.5.1.1). URL: <https://pubs.aeaweb.org/doi/10.1257/mac.5.1.1> (visited on 04/17/2026).
- Krusell, Per and Anthony A. Smith Jr. (Oct. 1998). “Income and Wealth Heterogeneity in the Macroeconomy”. en. In: *Journal of Political Economy* 106.5, pp. 867–896. ISSN: 0022-3808,

1537-534X. DOI: [10.1086/250034](https://doi.org/10.1086/250034). URL: <https://www.journals.uchicago.edu/doi/10.1086/250034> (visited on 10/10/2025).

Lee, Hanbaek (2025). “Global non-linear solutions in sequence space and the generalised transition function”.